The Dimension

Artist’s Proof 10

Dimensionality

Why three spatial dimensions — from four axioms

Contents

Artist's Note

This is Artist's Proof 10 of The 420 Code. The paper derives the number of spatial dimensions from the four axioms of the record algebra. It is the third chapter of Notebook II — Spacetime, following AP03 (The Ratio) and AP05 (The Break).

N = 3 is not an empirical input.

It is a consequence of the axioms.

The derivation proceeds in five structural steps. Each axiom is shown to express one independent face of the manifold (section 2). The independence of the faces follows from the proven independence of the axioms via the proven faithful embedding (section 3). The temporal versus spatial character of each direction is identified from the axiom's structural role (section 4). The result N = 3 is structurally consistent with the multidimensional residual — six faces pairing into the axis structure of 3+1 spacetime (section 5). The fifth degree of freedom — the 1:1 itself — is identified as the probability dimension (the Hilbert space), not a spatial dimension (section 7). This establishes the sufficiency of {S, B, R, C} to generate the record algebra: no fifth axiom is needed. The structural exhaustiveness claim — that no fifth operation on the 1:1 is *possible* — is held as a positive enumeration argument in section 7, contingent on the 1:1 being correctly identified as the substrate (not an operation) and on the four operations being correctly identified as cardinal.

AP10 closes three kill switches and leaves three live. KS-2c (CLOSED — contingent on KS-D.3): N = 3 derived from the four independent axioms via the proven faithful embedding, with the load-bearing one-axiom-one-face identity tested by KS-D.3. KS-15/D.2 (CLOSED — contingent on KS-D.4): axiom-to-dimension assignment unique — R is the only irreversible axiom; the formal signature derivation is fenced at KS-D.4. KS-16 (CLOSED — contingent on KS-P.1; sufficiency of {S, B, R, C}): the fifth degree of freedom is the 1:1 itself, identified as the Hilbert space; no fifth axiom is *needed*. The stronger claim that no fifth axiom is *possible* rests on section 7's positive enumeration of operations on the 1:1; that enumeration is held as a structural argument, not a closed theorem. KS-D.1 (LIVE — STRUCTURAL): the six-face count depends on AP06 section 10.5's residual structure. KS-D.3 (LIVE — STRUCTURAL): each axiom expresses exactly one face. KS-D.4 (LIVE — STRUCTURAL): the formal metric-level derivation of the Lorentzian signature; section 4 carries the structural argument.

AP10 closes the central conditional in AP08's derivation of Einstein's field equations. With N = 3 derived, Lovelock's theorem applies in 3+1 dimensions and the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$. Einstein's field equations with cosmological constant are an unconditional theorem of the record algebra.

AP10 is logically independent of AP01 and AP04 in the strict sense that its postulates (the four canonical axioms {S, B, R, C} and the proven hypotheses EH and QRA) are self-contained. AP10 *depends on* AP20 (which proves EH and QRA), AP06 (for the multidimensional residual confirming N = 3), and AP08 (for Lovelock's theorem, which AP10 in turn renders unconditional). The chain is bidirectional with AP08: AP08 establishes the conditional Lovelock derivation; AP10 removes the condition.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper is the third chapter of Notebook II — Spacetime. The Notebook II chapter sequence runs AP03 (The Ratio) → AP05 (The Break) → AP10 (this paper) → AP08 (The Identity).

Notebook II derives the structure of space, time, gravity, and the dimensional count from the axiom. AP03 establishes the conjugacy of c and G as outputs of ε. AP05 walks the axiom forward through the seven-step chain from symmetry to obligation, and reformulates the cosmological identity as Israel junction conditions on Σ. AP10 picks up from there: with the manifold installed (AP01, AP20) and the constants installed (AP03), AP10 pins the dimensional count. AP08 closes Notebook II by deriving the Einstein field equations on the dimensionally pinned manifold.

AP10 sits third in Notebook II because the dimensional count operates on the manifold AP01 and AP20 establish and the substrate AP03 derives. AP10 also closes the central conditional in AP08's field-equation derivation — AP08 derives Einstein's equations conditional on $N = 3$ via Lovelock's theorem; AP10 removes the condition by deriving $N = 3$ from the axioms. With AP10 in place, AP08's derivation is unconditional, and Notebook II reads as a closed structural chain from axiom to field equations.

AP10 is logically independent of AP01 and AP04 in the sense that its own postulates (the four canonical axioms {S, B, R, C}, the proven Embedding Hypothesis, and the proven Quantum-Record Alignment) are self-contained. AP10 depends on AP01 only for the formal apparatus of the record algebra; if AP01 stands, AP10's argument runs cleanly; if AP01 fails, the axioms fall back to declared assumptions, and AP10's argument runs against those assumptions instead.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Section 1 states the four axioms and the two proven hypotheses, and explains what was left undetermined. Sections 2 through 5 carry the structural argument: the four axioms express four independent faces of one manifold; the independence of the faces follows from the proven independence of the axioms; the temporal versus spatial character of each direction is identified; and the six-face residual structure provides an independent confirmation of N = 3. Section 6 names the consequences. Section 7 identifies the fifth degree of freedom as the Hilbert space. Section 8 is the falsification register. Section 9 is the synthesis.

Two voices alternate as the material requires. Narrative passages — the four axioms walked one at a time, the dimensions felt as they accumulate, the closing on the canvas and the painting — speak directly. Formal sections — the axiom statements, the independence theorems, the kill switches — speak in the precision the structure requires.

0 — Dependency and Scope

AP10 is the third chapter of Notebook II. Before the body, this section names what AP10 does, what it depends on, where its inputs map onto the canonical axioms {S, B, R, C}, what epistemic status each section holds, what kill switches are engaged, what structural debts remain, and how AP10 stands in relation to the rest of The 420 Code.

0.1 — What this paper does

AP10 derives N = 3 spatial dimensions from the four axioms of the record algebra. The argument is structural and runs in one primary form, with a confirming consistency check from a different feature of the architecture. The primary form (sections 2 through 4) identifies each of the four axioms with one independent face of the embedded manifold; the independence of the faces follows from the proven independence of the axioms via the proven faithful embedding; the temporal versus spatial character of each direction is identified from the axiom's structural role. The confirming form (section 5) counts the six faces of the multidimensional residual (AP06 section 10.5) and shows they pair naturally into the axis structure of 3+1 spacetime — two purely spatial axes (G–c and α/β–α_em) and one spatial-temporal connection axis (m_e–t). The residual structure is structurally consistent with the primary derivation of 3+1; it does not constitute an independent derivation of N = 3 spatial, because Pair 3 includes the temporal face t. Section 5 is therefore a structural consistency check via a different feature of the architecture, not a doubly-independent derivation.

AP10 also identifies the fifth structural degree of freedom (section 7) as the 1:1 itself — the pre-state from which {S, B, R, C} operate — and locates it as the Hilbert space (the probability dimension), not a fifth spatial dimension. This closes KS-16 in the sufficiency sense: {S, B, R, C} suffice to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1); the 1:1 itself is structurally prior to the manifold rather than a feature of it. The stronger exhaustiveness claim — that no fifth independent operation on the 1:1 is *possible* — is held in section 7 as a positive enumeration of the operations performed on the pre-state, with the closure resting on the 1:1 being correctly identified as the substrate (not an operation) and on the four operations being correctly identified as the cardinal independent operations on it.

The contributions are three kill-switch closures (KS-2c, KS-15/D.2, KS-16) and three live kill switches (KS-D.1, KS-D.3, KS-D.4) targeting the load-bearing identifications the argument relies on. AP10 also closes the conditional in AP08's derivation of Einstein's field equations: AP08 derives the equations conditional on N = 3 via Lovelock's theorem; AP10 derives N = 3 from the axioms; the conditional is removed; Einstein's field equations with cosmological constant become an unconditional theorem of the record algebra.

0.2 — Dependencies

AP10 is a structural-derivation paper. Its dependencies are explicit and listed here in compact form so the load-bearing imports are visible at the door.

AP20 (the canonical axioms). Load-bearing for the four canonical axioms {S, B, R, C} (formal statement at AP20 section 3.3) and for their independence (the four removal arguments at AP20 section 4.2 — stated results with proof sketches; full formalisation fenced at KS-P.2). The axioms are jointly exhibited by any existing record (the undeniable premise, AP20 section 6), which carries their mutual consistency. The independence arguments are the foundation of section 3's argument: each axiom is independent of the other three; therefore the face expressed by each axiom on the manifold is independent of the faces expressed by the other three; therefore the four faces are independent dimensions.

The Lorentzian signature (section 4, KS-D.4). Load-bearing for the signature (−, +, ..., +) and, with it, the symmetry group SO(1, N) (the isometry group of that signature). The signature is carried in this paper: section 4's structural argument reads it from the axioms themselves — exactly one axiom (R) is intrinsically irreversible, and irreversibility read as metric character is the (−) sign. The formal metric-level derivation is the open obligation fenced at KS-D.4. AP10 pins the value of N.

AP20 (EH and QRA proven). Load-bearing for the faithful embedding of the record algebra into a smooth manifold (Embedding Hypothesis, EH — closed at AP20) and for the quantum-record identity (Quantum-Record Alignment, QRA — closed at AP20 section 5.5, KS-P.4). The differentiability and leading-order quadratic structure of the cone boundary used throughout rest on the smooth-manifold structure whose obligation is fenced at KS-P.5. EH is invoked at section 3.2 to convert algebraic independence into geometric face-independence.

AP06 section 10.5 (multidimensional residual). Load-bearing for the six-face structure of the break invoked in section 5. AP06's residual analysis identifies six faces — G, c, α/β, α_em, m_e, t — forming the structural inventory of what the break produces. AP10 reads this as three conjugate pairs spanning three spatial axes, providing an independent confirmation of N = 3. KS-D.1 is the kill switch on AP10's reading of AP06's residual structure.

AP08 section 9 (Lovelock's theorem, formerly conditional on N = 3). Load-bearing structurally but inverted in direction: AP08 derives Einstein's field equations from Lovelock's theorem conditional on N = 3; AP10 derives N = 3 and removes AP08's condition. The relationship is bidirectional rather than purely upstream: AP10 depends on AP08's derivation existing; AP08 depends on AP10's N = 3 for unconditionality.

AP19 sections 2–3 (three faces of one manifold). Load-bearing for the structural reading of the spatial faces as three faces of a single manifold (not three separately assembled pieces). AP10's section 2.5 (Co-arising) inherits the formulation from AP19.

Internal cross-references to AP09 (the Hilbert space as pre-state from which quantum mechanics operates), AP12 ($\hbar$ identification), and AP24 (m_e as residual of the break) are noted at the relevant sections but are not load-bearing for AP10's derivation.

0.3 — Axiom mapping

AP10 was written after AP20's Paper D formalised the four canonical axioms {S, B, R, C}, and it is one of the papers in which the mapping from axioms to structural content is most direct. The mapping here is not retrospective; it is the paper's subject.

Axiom S (Symmetry) ↔ The exchange face. Two disjoint sectors $\mathcal{C}$ and $\mathcal{P}$ with order-reversing involution σ. On the embedded manifold, σ acts as a $\mathbb{Z}_2$ symmetry across one direction; that direction is the exchange face. Spatial. Signature: (+).

Axiom B (Unique Breaking) ↔ The break face. One element $\varepsilon \in \mathcal{C}$ with no σ-image; valuation $\nu(\mathcal{C}) - \nu(\boldsymbol{\mathcal{P}}) = \nu(\varepsilon) = 1$. On the manifold, ε is a distinguished locus with a direction of advance; that direction is the break face. Spatial. Signature: (+).

Axiom R (Record Monotonicity) ↔ Time. Sequential composition forms a monoid (no inverse); records cannot be locally annihilated; history is irreversible. On the manifold, this produces a distinguished irreversible direction; that direction is time. Signature: (−).

Axiom C (Constraint) ↔ The propagation face. Finite invariant rate c constraining sequential propagation, structural rather than electromagnetic. On the manifold, c produces a light cone with a definite opening angle; the direction along which propagation is maximally extended is the propagation face. Spatial. Signature: (+).

The 1:1 (pre-state) ↔ The Hilbert space. The state on which {S, B, R, C} operate. Not a feature of the record algebra's structure on the manifold but the state the algebra breaks *from*. On the manifold, the 1:1 appears as the Hilbert space — the probability dimension perpendicular to spacetime. Pre-spatial and pre-temporal. Not a fifth manifold direction.

The four axioms map one-to-one onto the four faces of the manifold; the 1:1 maps onto the Hilbert space (the probability dimension) which is prior to the manifold. Five structural features total; four on the manifold and one prior to it.

0.4 — Epistemic status per section

Each section carries an explicit status — derivation, structural argument, formal block, or synthesis — so the reader knows what kind of weight each carries.

Section 1 (The Starting Point). FOUNDATIONAL. The four axioms stated; the proven hypotheses (EH and QRA) named; the undetermined quantity (N) flagged.

Section 2 (Four Axioms, Four Faces). STRUCTURAL ARGUMENT. Each axiom identified with one independent face of the embedded manifold; the faces named as co-arising rather than separately assembled.

Section 3 (Independence). DERIVATION. The independence of the four axioms (the removal arguments at AP20 section 4.2, fenced at KS-P.2) combined with the proven faithful embedding (AP20) yields the independence of the four geometric faces; four independent faces yield four dimensions; N = 3.

Section 4 (Character of Each Direction). DERIVATION. The temporal versus spatial character of each direction identified from the axiom's structural role: R alone is irreversible and therefore gives the (−) direction; the other three axioms are compatible with reversibility in their directions and therefore give the (+) directions.

Section 5 (The Six Faces). INDEPENDENT CONFIRMATION. The six faces of the multidimensional residual (AP06 section 10.5) read as three conjugate pairs spanning three spatial axes. Independent of section 2–3's axiom-count argument; arrives at the same N = 3.

Section 6 (Consequences). SYNTHESIS + CLOSURE. KS-2c closed. Lovelock's theorem (AP08 section 9) becomes unconditional. The 420 Code's chain from axioms to Einstein's field equations is closed.

Section 7 (The Fifth Degree of Freedom). STRUCTURAL ARGUMENT + CLOSURE. The 1:1 itself identified as the fifth structural degree of freedom — the Hilbert space, the probability dimension, the pre-state. Not a fifth spatial dimension because pre-spatial. KS-16 closed.

Section 8 (Kill Switches). FORMAL FALSIFICATION REGISTER. Three closed (KS-2c, KS-15/D.2, KS-16 — each carrying its named contingency) and three live (KS-D.1, KS-D.3, KS-D.4).

Section 9 (What This Establishes and What Remains Open). SYNTHESIS.

0.5 — Kill switch summary

Six kill switches are engaged in this paper. Three are closed by AP10; three remain live. The full register is in section 8.

KS-2c: N = 3 spatial dimensions. CLOSED — derived in sections 2–3 via the proven faithful embedding. The closure is contingent on KS-D.3 (one axiom, one face) remaining unfired; if KS-D.3 fires, KS-2c reopens. Section 5 provides structural consistency from the residual face count, not an independent derivation.

KS-15/D.2: Axiom-to-dimension assignment unique. CLOSED — contingent on KS-D.4. R is the only irreversible axiom; the assignment R → time, {C, S, B} → space is forced once the Lorentzian signature holds (section 4's structural argument; formal derivation fenced at KS-D.4).

KS-16: Sufficiency of {S, B, R, C}. CLOSED — contingent on KS-P.1. The four canonical axioms suffice to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1); the fifth structural degree of freedom is the 1:1 itself, identified as the Hilbert space; no fifth axiom is needed. The exhaustiveness claim (no fifth axiom is *possible*) is held as a positive enumeration in section 7, contingent on the four operations being correctly identified as cardinal.

KS-D.1: Six-face count and axis-structure interpretation. LIVE — STRUCTURAL. The section 5 argument depends on the identification of the six residual faces from AP06 section 10.5 *and* on their reading as pairs that match the axis structure of 3+1 spacetime. If the residual has a different face count, or the pairs do not correspond to the expected axis structure (two spatial axes plus one spatial-temporal connection), the section 5 consistency check fails. The primary argument (sections 2–3) survives.

KS-D.3: One axiom, one face. LIVE — STRUCTURAL. The section 2 argument rests on the identity *one independent algebraic axiom = exactly one independent geometric face* (via the proven faithful embedding, AP20). The faithful embedding gives distinct-to-distinct (injectivity); the load-bearing additional claim is that the embedding is dimension-minimal: each independent axiom contributes exactly one new direction, not zero (acting as a constraint within existing dimensions) and not two (contributing multiple independent directions). If KS-D.3 fires at any axiom, the count $N = 3$ is wrong.

KS-D.4: Signature formalisation. LIVE — STRUCTURAL. Section 4 reads the Lorentzian signature from the axioms' own reversibility structure — exactly one axiom (R) is intrinsically irreversible, and the irreversible direction carries the ($-$) metric character. The formal metric-level derivation is the open obligation this switch names; KS-15/D.2's closure is contingent on it.

0.6 — Structural debts owed

Three structural-class items are tied to AP10's argument. None is a registry-level debt in the same sense as AP05's Open Problem B-JP; they are the load-bearing structural identifications that the three live kill switches track. Named here so the structural exposure is visible at the door.

Six-face structure of the break (KS-D.1). The argument in section 5 reads AP06 section 10.5's residual analysis as identifying exactly six faces of the break, pairing into three conjugate pairs. If a more careful or extended analysis of the residual produces a different face count — four, eight, or a non-pair structure — section 5's confirmation collapses. The primary argument (sections 2–3) is independent of the face count and survives such a failure. AP10 holds KS-D.1 as a live structural test of the section 5 confirmation, not of the central argument.

One axiom expresses exactly one face (KS-D.3). The argument in section 2 reads each axiom as expressing exactly one face of the embedded manifold. The identity rests on the proven faithful embedding (AP20): if EH is faithful, then independent algebraic content maps to independent geometric content, and the four independent axioms map to four independent geometric features. The identity could fail in two ways: an axiom that expresses zero geometric content (acting only as a constraint on existing structure) or an axiom that expresses two independent geometric directions (contributing more than one new face). AP10 holds KS-D.3 as a live structural test of the central argument.

The Lorentzian signature's formal derivation (KS-D.4). Section 4 reads the signature from the axioms' own reversibility structure: R is the only intrinsically irreversible axiom, and the irreversible direction carries the (−) metric character. The formal metric-level derivation — that the embedded geometry is pseudo-Riemannian with exactly this signature — is the open obligation KS-D.4 names. AP10 holds KS-D.4 as a live structural test of the section 4 distinction.

0.7 — Items held open without claiming debt status

One item is bracketed inside this paper.

The mapping from face to specific geometric coordinate. AP10's argument shows that there are four independent faces and that one is temporal and three are spatial. It does not specify which coordinate within the spatial three-space is "the C direction" versus "the S direction" versus "the B direction" — the three spatial axes are interchangeable under spatial rotation (SO(3) symmetry of the spatial manifold). The argument fixes the count and the signature; it does not fix a coordinate-level identification. This is not a structural debt because the rotational symmetry of three-space is the expected and correct outcome; held without claiming a separate debt.

0.8 — Structural relationships

AP10 is the third chapter of Notebook II. The chapter sequence runs AP03 → AP05 → AP10 → AP08. AP03 establishes the constants (c, G) as outputs of ε on the substrate; AP05 walks the axiom forward through the seven-step chain to obligation and reformulates the cosmological identity as junction conditions; AP10 pins N = 3 on the manifold the chain occupies; AP08 closes Notebook II by deriving Einstein's field equations on the dimensionally pinned manifold.

AP01 (Notebook I) is the source of the record algebra. AP20 (Notebook I) is the source of the Embedding Hypothesis proof and the Quantum-Record Alignment proof, and of the stated-and-fenced completeness and minimality results for the axiom set (AP20 section 4; KS-P.1/KS-P.2). AP06 (Notebook IV) is the source of the multidimensional residual analysis used in section 5. AP08 (Notebook II) is the source of Lovelock's theorem in the body of work context, and is the paper whose central conditional AP10 closes. AP19 (Notebook VI) is the source of the "three faces of one manifold" reading inherited by section 2.5.

Downstream, AP10 feeds AP08 (closing the N = 3 conditional for Einstein's field equations), AP09 (the Hilbert space identification at section 7 connects to AP09's pre-state formulation of quantum mechanics), AP12 ($\hbar$ identification operates on the Hilbert space identified here), AP24 (m_e identification operates within the residual structure used in section 5), and the entire downstream cosmological and quantum programme that assumes N = 3 unconditionally.

1 — The Starting Point

1.1 — The four axioms

You have four axioms.

AP20 Paper D installed them. The 420 Code has been using them.

Look at them now as the floor of what is about to happen.

S (Symmetry). Two disjoint sectors $\mathcal{C}$ and $\boldsymbol{\mathcal{P}}$ with order-reversing involution σ. Extensive quantities match: Q($\mathcal{C}$) = Q($\boldsymbol{\mathcal{P}}$) in the unbroken case.

B (Unique Breaking). One element ε ∈ $\mathcal{C}$ with no σ-image. Valuation: $\nu(\mathcal{C}) - \nu(\boldsymbol{\mathcal{P}}) = \nu(\varepsilon) = 1$. This is the break. Without B, the system is the 1:1 — perfectly symmetric, and nothing exists.

R (Record Monotonicity). Sequential composition (·) forms a monoid, not a group, within each sector. No non-identity element has an inverse. Records cannot be annihilated locally. History is irreversible.

C (Constraint). Finite invariant rate c constraining sequential propagation. Structural, not electromagnetic.

These four axioms are independent (the four removal arguments at AP20 section 4.2; full formalisation fenced at KS-P.2) and jointly consistent — any existing record exhibits all four jointly satisfied (the undeniable premise, AP20 section 6).

Independence means no axiom is derivable from the other three. Removing any one axiom produces a strictly weaker system. Each axiom adds irreducible content to the algebra.

1.2 — What is now proven

Two hypotheses were carried as conditionals in earlier APs. Both are now proven (AP20).

EH (Embedding Hypothesis). The algebraic pre-state structure defined by {S, B, R, C} embeds into physical reality as a smooth manifold M. Proven in AP20.

QRA (Quantum-Record Alignment). Quantum states are pre-state records. Closed at AP20 section 5.5 (KS-P.4). The differentiability and leading-order quadratic structure of the cone boundary invoked here rest on the smooth-manifold structure fenced at KS-P.5.

From the axioms, the record algebra produces a Lorentzian manifold (M, g) with signature (−, +, …, +) and symmetry group SO(1, N).

1.3 — What was left undetermined

The number of + signs — the number of spatial dimensions N — is not fixed by the signature. The signature argument (section 4; formal derivation fenced at KS-D.4) establishes that the manifold is Lorentzian and that the symmetry group is SO(1, N). It does not fix N.

This paper fixes N.

Specifically: N is determined by the number of independent axioms, with a structural consistency check from the multidimensional residual. A primary derivation — the axiom count — fixes N = 3; the residual face structure confirms it. The two are not doubly independent: the residual includes the temporal face, so it is a cross-check, not a second derivation.

2 — Four Axioms, Four Faces of One Manifold

Each axiom contributes structure to the embedded manifold.

But the manifold is one structure — not four separate pieces assembled together.

The axioms express *faces* of this structure. Each axiom names one irreducible feature that the manifold must have.

The faces are intrinsically linked — they co-arise in every actualisation event — but distinct, because no axiom is derivable from the others.

The claim is short.

Four independent axioms express four independent faces. Four faces, four dimensions.

2.1 — R → Time

Axiom R states: records accumulate irreversibly.

The monoid has no inverse. History cannot be undone.

On the embedded manifold, this produces a distinguished direction.

The direction in which records accumulate. The direction of the now. The direction of actualisation.

This is the time direction.

The irreversibility of R gives the direction its character: opposite sign to the spatial directions (Lorentzian signature), cannot be reversed (arrow of time), the direction along which the monoid composition operates.

You have just watched the first dimension emerge.

R expresses the temporal face: time. Signature: (−).

It is the only irreversible axiom — the only face with a preferred direction (forward, never backward).

This is the (−) in the Lorentzian signature.

The remaining three axioms — C, S, B — express the three spatial faces. These faces are intrinsically linked: they co-arise in every actualisation event, because every record requires propagation (C), sector structure (S), and a break (B) simultaneously. You cannot have one without the others. But they are distinct: no axiom is derivable from the others.

Linked but not reducible. Three faces of one manifold.

2.2 — C → The Propagation Face

Axiom C is Constraint — locality: a record is somewhere, not everywhere. On the manifold, the Constraint produces a finite invariant rate c constraining sequential propagation.

Imagine C removed. What you have is a manifold where the record written at point x reaches point y instantly.

Every point causally connected to every other point at every time.

Spatial separation has no physical meaning.

You have a time direction (from R) but no spatial extension. A one-dimensional manifold. A line.

Time without anywhere to go.

C creates spatial extension. It says: the record written at point x cannot influence point y instantly. There is a finite delay proportional to the separation between x and y.

This delay creates distance.

The separation between causally disconnected events is what makes space spatial.

On the manifold, C produces the light cone — the boundary between events that can and cannot be causally connected from a given point. The light cone has a definite opening angle (determined by c).

The direction along which propagation is maximally extended is a spatial direction. It is the propagation direction.

The direction is independent from R. Without R, C has nothing to constrain. Without C, R produces no spatial structure.

Together, R and C produce 1+1 dimensions: time and one spatial direction. The light cone in its minimal form.

C expresses one spatial face: propagation. Signature: (+).

2.3 — S → The Exchange Face

Axiom S states: two disjoint sectors $\mathcal{C}$ and $\boldsymbol{\mathcal{P}}$ with order-reversing involution σ.

On the manifold, the involution σ acts as a $\mathbb{Z}_2$ symmetry.

The direction in which σ acts is a direction on the manifold.

It is the direction of crossing between sectors.

The direction is independent from both R and C.

R gives the direction of time. C gives the direction of propagation. S gives a third direction: the direction across which the two sectors are distinguished.

Consider two records, one in $\mathcal{C}$ and one in $\boldsymbol{\mathcal{P}}$, at the same time (R) and the same propagation distance (C). They still differ — they are in different sectors.

The direction of their difference is neither temporal nor propagational.

A third direction: the sector direction. The involution σ acts along this direction, mapping one to the other.

Without S, the manifold would have at most two dimensions.

No “width” — no direction perpendicular to both the time direction and the propagation direction.

S creates this width.

S expresses one spatial face: exchange. Signature: (+).

Three dimensions. You can feel the fourth coming.

2.4 — B → The Break Face

Axiom B states: one element $\varepsilon \in \mathcal{C}$ with no σ-image in $\boldsymbol{\mathcal{P}}$.

On the manifold, ε is a distinguished locus — the point where the symmetry between $\mathcal{C}$ and $\boldsymbol{\mathcal{P}}$ is broken.

The break occurs at a specific *location* in the manifold. The location is a spatial fact — a "where," not a "when." The temporal aspect of the break (records accumulating irreversibly as the break propagates) is carried by R, not by B. B contributes the spatial direction associated with the location; R contributes the temporal direction along which the location's consequences accumulate.

The direction is independent from R, C, and S.

R says *when*. C says *how fast*. S says *between what*. B says *where* — which degree of freedom will be broken next, in space.

This *where* is a spatial direction independent of the other three.

Consider the now (the uncoupled ε) at a given time (R), propagating at a given speed (bounded by C), in the space between the two sectors (S). The spatial location at which the now is found is not determined by R, C, or S alone.

B gives that location.

Without B, the manifold would have at most three dimensions. No "depth" — no direction corresponding to the specific location of the break in the space of possibilities.

B creates this depth by placing ε at a specific point in the manifold.

A note on B and the faithful embedding. B is algebraically independent of {R, C, S} (the removal argument for B at AP20 section 4.2; fenced at KS-P.2); by the proven faithful embedding, B's independent algebraic content maps to independent geometric content on the manifold. The load-bearing additional claim — that B's geometric content takes the form of a new spatial direction rather than a constraint within R+C+S — is the central identity tested by KS-D.3. The reading made here (B → break face, signature +) is the reading consistent with the dimension-minimal embedding; KS-D.3 is the live structural test of that reading.

B expresses one spatial face: the break direction. Signature: (+).

Four dimensions. The count is complete.

2.5 — Co-arising of the four faces

One temporal face (R) and three spatial faces (C, S, B).

These are not four separate pieces assembled together. They are four expressions of one manifold, co-arising in every actualisation event.

The manifold does not exist first and then receive the axioms. The axioms and the manifold co-arise.

The break is the event. The manifold is the description of the event. The faces are the structure of the description.

The number of independent faces IS the dimensionality.

3 — Independence of Faces from Independence of Axioms

3.1 — The theorem

AP20 section 4.2 carries the four removal arguments — stated results with proof sketches, full formalisation fenced at KS-P.2: each axiom is independent of the other three.

Removing any one axiom produces a strictly weaker system.

No axiom is derivable from the others.

3.2 — The consequence

If axiom X is independent of axioms {Y, Z, W}, then the face expressed by X cannot be a combination of the faces expressed by {Y, Z, W}.

If it could, then X’s structural content on the manifold would be derivable from {Y, Z, W}’s content — which would mean X is derivable.

But EH is proven (AP20).

The embedding is faithful. Structure in the algebra maps to structure on the manifold. Distinct algebraic content maps to distinct geometric content.

Therefore X’s independence in the algebra entails X’s face-independence on the manifold.

Independent faces express independent dimensions.

3.3 — The result

Four independent axioms → four independent faces of one manifold → four dimensions.

R gives one temporal direction: (−).

C, S, B give three spatial directions: (+, +, +).

Signature: (−, +, +, +). Dimension: 3+1.

N = 3.

Not a coincidence.

The number of spatial dimensions IS the number of independent spatial faces of the manifold. And the number of independent spatial faces is the number of independent axioms minus the one that gives time.

3.4 — Why not more, why not fewer

Why not N > 3?

To get a fifth dimension, you would need a fifth independent axiom.

But the record algebra is completely specified by {S, B, R, C}. The completeness result stated at AP20 section 4.1 (proof sketch carried there; full formalisation fenced at KS-P.1) is that these four suffice to generate the full algebra. No fifth axiom is needed.

The four axioms already cover: symmetry (S), breaking (B), irreversibility (R), and bounding (C).

What structural feature is missing? What could a fifth axiom say that {S, B, R, C} do not already determine?

The algebra of records — symmetric sectors, one break, irreversible composition, finite propagation — is complete.

There is no room for a fifth independent structural feature.

Why not N < 3?

AP20 section 4.2 carries the four removal arguments: each axiom is independent of the other three (full formalisation fenced at KS-P.2).

Removing any one produces a strictly weaker system.

With only R and C (no S, no B), you get 1+1 dimensions: a line with a speed limit. With R, C, and S (no B), you get 1+2 dimensions: a sheet with sector structure but no break. The full 1+3 dimensions require all four axioms.

$N < 3$ means an axiom is missing.

The number of spatial dimensions is the number of independent axioms minus the one that gives time.

You are looking at the answer.

It was in the axioms from the start.

4 — The Character of Each Direction

One direction has opposite metric sign to the others — the Lorentzian signature. Its structural source is in the axioms themselves: exactly one axiom is intrinsically irreversible, and irreversibility read as metric character is the (−) sign; the formal metric-level derivation is the open obligation fenced at KS-D.4.

This section identifies which axiom produces which character.

4.1 — Time is R

The (−) direction is the one in which records accumulate.

R is the only axiom that introduces irreversibility — the monoid has no inverse.

All other axioms are compatible with reversibility in their contributed direction.

C gives a symmetric speed limit (propagation at c in either spatial direction).

S gives a symmetric involution (σ maps $\mathcal{C} \to \boldsymbol{\mathcal{P}}$ and $\boldsymbol{\mathcal{P}} \to \mathcal{C}$ equally).

B gives a definite break direction but does not prohibit the reverse spatial direction.

You cannot go backward in time because Axiom R has no inverse.

Only R contributes a direction that is intrinsically asymmetric — that has a “forward” but no “backward.”

The (−) in the signature.

4.2 — Space is {C, S, B}

The three (+) directions are the ones in which records can exist at either end — where forward and backward are structurally symmetric.

Spatial directions admit motion in both directions.

Temporal direction does not admit reversal of records.

This is the signature distinction.

C gives propagation distance: symmetric. You can travel in either direction along the propagation axis, up to speed c.

S gives sector-crossing: symmetric. σ maps in both directions by definition — it is an involution, σ^2 = identity.

B gives the break direction: the now advances in a specific direction, but the spatial axis itself admits both directions. The asymmetry of the break is *temporal* (the now writes records irreversibly, via R), not spatial.

The signature $(-, +, +, +)$ is the axiom structure (R, C, S, B) read as metric character.

5 — The Six Faces

A structural consistency check on the dimensional count via the multidimensional residual.

AP06 section 10.5 identifies the multidimensional residual of the break: one break, six faces.

The six faces are: G (geometry/curvature), c (propagation bound), α/β (substrate stiffnesses), $\alpha_{em} \approx 1/137$ (electromagnetic coupling), m_e (what escaped), and t (time direction).

The six faces pair into three conjugate pairs. Two of the three pairs are purely spatial axes; the third pair connects spatial to temporal. The structure is consistent with the 3+1 count derived in sections 2–3 (three spatial axes + one temporal direction), and the consistency provides a structural cross-check from a different feature of the architecture (the residual) rather than the axiom count.

A note before the pairs are walked. The kind of conjugacy each pair carries is not strictly homogeneous: Pair 1 (G–c) and Pair 2 (α/β–α_{em}) are intrinsic-property splits within the spatial structure; Pair 3 (m_e–t) is a spatial-temporal split that connects the spatial content of the break to the temporal direction along which records accumulate. The three pairs share the property that the residual structure pairs naturally; they differ in whether the pairing is internal to the spatial axes or between the spatial and the temporal.

5.1 — Six faces, three pairs

Pair 1: G and c. *Purely spatial axis.*

The two curves of the eye.

G measures curvature — how tightly the condensate folds.

c measures propagation — how fast the break can travel.

They are conjugate: G determines what happens when records accumulate maximally, c determines what happens when records propagate maximally. One is the spatial response to density. The other is the spatial response to speed.

Together they span one spatial axis.

Pair 2: α/β and α_em. *Purely spatial axis.*

The internal stiffnesses and the coupling constant.

α and β are substrate stiffnesses ($c^2 = \beta/\alpha$).

$\alpha_{em} \approx 1/137$ is the fine-structure constant.

The stiffnesses determine the fabric's internal response. The coupling constant determines how strongly ε interacts with the field.

Conjugate: α/β is resistance to deformation, α_em is willingness to break.

Together they span one spatial axis.

Pair 3: m_e and t. *Spatial-temporal connection axis.*

What escaped and in which direction.

m_e is the electron mass — the mass of ε, the residual of the break.

t is the time direction — the direction in which records accumulate.

m_e is the spatial content of the break (the mass of the residual exists in space). t is the temporal content (the direction along which the break's consequences propagate forward in time).

Pair 3 is not a purely spatial axis. It is the connection between the spatial residual (m_e) and the temporal direction (t) along which the residual's history accumulates. The primary derivation already sources time from R and space from {C, S, B} separately; Pair 3 in the residual structure reads as the visible connection between those two layers, not as a third spatial axis.

5.2 — What the pairing shows

The six faces of the multidimensional residual pair into the axis structure of 3+1 spacetime: two purely spatial axes (G–c, α/β–α_em) and one spatial-temporal connection axis (m_e–t).

This is consistent with 3+1, where the manifold has three independent spatial axes and one temporal direction. The two purely spatial residual pairs correspond to two of the three spatial axes; the third spatial axis is not separately exhibited by the residual structure (which is one of the limits of this consistency check); the spatial-temporal pair shows that the residual is a coherent connection between space and time, not a mismatched count.

A direct alternative reading is also available: each of the three pairs spans one of the three axes of *the spatial manifold plus its temporal connection*, with the temporal direction visible as one end of one of the three pairs. Under this reading, the six faces are the six face-directions of 3+1 spacetime (three spatial axes × two ends + the temporal direction explicit in Pair 3). Either reading yields the same conclusion: the residual structure is consistent with 3+1.

What this is NOT. It is not an independent derivation that N = 3 spatial. The temporal face t is explicitly one of the six residual faces, so the six-face count includes both spatial and temporal content; reading "six = three spatial axes × two ends" without accounting for the temporal face would double-count time. The primary derivation in sections 2–3 sources time and space separately and counts only the spatial dimensions; the residual structure here is a consistency check on the total 3+1 architecture, not on the spatial count in isolation.

5.3 — Structural cross-check, not independent derivation

The argument in sections 2–4 derives N = 3 spatial from the number and independence of the axioms.

The argument in section 5.1–5.2 checks structural consistency by counting the residual faces and matching them against the axis structure of 3+1 spacetime.

The two use different features of the architecture (axiom count vs residual count) and arrive at compatible results.

The convergence is a structural consistency check rather than a doubly-independent derivation. If the residual count had been incompatible with 3+1, the architecture would have a contradiction; it does not.

The consistency check is necessary but not sufficient: a different residual structure could in principle have been consistent with N = 3 too, and AP06's six-face count is itself a structural identification rather than a forced output of the axioms. Section 5 tells us the residual structure does not contradict the primary derivation. The primary derivation alone closes KS-2c; section 5 is the cross-check from a different feature of the architecture.

6 — Consequences

6.1 — KS-2c is closed

Kill switch KS-2c asked: why N = 3 spatial dimensions?

This paper answers: because there are four independent axioms, one gives the temporal direction, three give the spatial directions.

N = 3 is derived, not assumed.

KS-2c is closed.

6.2 — Lovelock is unconditional

AP08 section 9 derived Einstein's field equations via Lovelock's theorem, conditional on N = 3.

With N = 3 now derived, the condition is removed.

Lovelock's theorem (Lovelock 1971) applies in all dimensions; the *uniqueness* in 3+1 specifically — the divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is uniquely $G_{\mu\nu} + \Lambda g_{\mu\nu}$ — is the Lovelock 1972 result. In other dimensions, the general theorem admits additional terms in the Euler density expansion; in 3+1 specifically those terms are topological or vanish, leaving only the Einstein tensor and the cosmological constant.

Einstein's field equations with cosmological constant are now an unconditional theorem of the record algebra.

The derivation chain is complete:

Axioms {S, B, R, C} → independent (AP20 section 4.2, fenced at KS-P.2) and jointly exhibited by any existing record (AP20 section 6).

+ EH + QRA (proven, AP20) → Lorentzian manifold (section 4's signature argument; fenced at KS-D.4).

+ Four independent axioms → four dimensions, signature (−, +, +, +). (This paper.)

+ Record density on M → Poisson forced by symmetry constraints (AP08 section 4).

+ Lovelock's theorem (now unconditional) → $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. (AP08 section 9.)

No link in this chain is conditional on an empirical input for the number of dimensions.

The form of Einstein's field equations is derived from the axioms alone.

6.3 — What this means for The 420 Code

After this paper, The 420 Code derives Lorentzian spacetime, special relativity, $N = 3$ spatial dimensions, the Poisson equation, Einstein's field equations, the cosmological constant existence, and quantum mechanics — all unconditional.

No empirical inputs other than m_e (from AP24) and the value of G (identified but not independently computed, per AP08 section 5).

EH and QRA are proven (AP20). All results are unconditional.

7 — The Fifth Degree of Freedom

7.1 — The question

The argument in sections 2–3 derives four dimensions from four axioms.

The natural challenge: can a fifth independent axiom be added to the record algebra?

If so, the architecture predicts a fifth dimension.

No fifth dimension is observed.

This was KS-16.

7.2 — The answer

There IS a fifth structural degree of freedom.

It is not missing. It is the foundation.

You have been standing on it the entire time.

The 1:1 itself — the pre-state, the @ AS prior, the state from which {S, B, R, C} operate — is a degree of freedom.

It has structure. It is not empty.

It contains the probability of every possibility. It is the space from which the break draws. It is the space to which defragmentation returns.

But the 1:1 does not produce a fifth *spatial* dimension.

It produces the Hilbert space — the space in which the wave function lives. The probability space. The amplitude space. The dimension perpendicular to spacetime.

7.3 — Why it is not on the manifold

The four axioms {S, B, R, C} act ON the 1:1. They produce the manifold.

The 1:1 is what the axioms act on — it is the stage, not an actor.

The actors (the axioms) produce four independent directions on the manifold.

The stage (the 1:1) is what the manifold is embedded in. It is prior to the manifold.

It does not appear as a dimension of the manifold for the same reason the canvas does not appear as a colour in the painting.

You do not see the canvas because you are painted on it.

The record algebra embeds into a smooth manifold (AP20). The manifold has the dimensions contributed by the axioms — four.

The 1:1 does not embed as a manifold direction because it is not a feature of the record algebra's structure on the manifold. It is the state the algebra breaks FROM.

On the manifold, the 1:1 appears as the Hilbert space.

The space of amplitudes. The space of possibilities. The space in which the wave function takes its values.

The probability dimension — the dimension that tells you not WHERE something is on the manifold, but HOW LIKELY it is to be found there when the now writes a record.

7.4 — The complete structure

Five structural features. Five degrees of freedom. But they are not all of the same kind.

The 1:1 → the probability dimension. Pre-spatial. Pre-temporal. The Hilbert space.

R → time. The irreversible direction. (−).

C → propagation. Spatial. (+).

S → sector-crossing. Spatial. (+).

B → break direction. Spatial. (+).

Four dimensions on the manifold: (−, +, +, +).

One dimension prior to the manifold: the probability space.

No fifth axiom is needed: {S, B, R, C} generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1). The sufficiency of the four axioms is stated at AP20 and fenced at KS-P.1.

The stronger claim that no fifth axiom is *possible* requires more than the sufficiency theorem. It requires a positive enumeration of the operations performed on the 1:1 and an argument that the four named operations exhaust the cardinal structural moves. The enumeration is the section’s thesis:

Symmetry (S) gives the 1:1 two sectors — the binary structural distinction that any actualised structure must support. Breaking (B) splits the symmetric pair — the single break that initiates the asymmetric instantiation. Recording (R) makes the split permanent — the irreversible accumulation that prevents return to the symmetric prior. Bounding (C) makes the propagation finite — the structural cap that gives the split a definite extent.

These four operations cover the four cardinal structural moves on a symmetric pre-state: distinguish, break, record, bound. The argument that no fifth operation exists rests on these four being cardinal — each addressing one independent structural question (distinguish? break? record? bound?) about the relationship between the pre-state and the resulting structure. A candidate fifth operation would have to address a fifth independent structural question; the four-question enumeration is the 420 Code’s claim that no such question remains.

This is a positive enumeration argument, not a closure theorem. It is held in section 8 as KS-16’s structural content. If a fifth cardinal structural question were identified, KS-16 would reopen. The closure as it stands rests on the four-question enumeration matching the structural moves required to convert a symmetric pre-state into an actualised structure.

What remains is the 1:1 itself — the probability space from which all actualisation draws.

Structurally complete. Not because we say so. Because the operations performed on the 1:1 by the four axioms address the four cardinal structural questions, and the 1:1 itself is not an operation but the substrate operations act on.

You are looking at the complete structure.

The universe has three spatial dimensions because the 1:1 has four operations performed on it, one of which is irreversible (giving time) and three of which are reversible (giving space).

There is no fifth spatial dimension because there is no fifth operation.

There is no fifth operation because the 1:1 is already fully addressed by the four cardinal structural questions: distinguish, break, record, bound.

And the 1:1 itself — the thing being broken — is the Hilbert space, the probability dimension, the wave function’s home.

It is the fifth degree of freedom.

It was always there.

It was never missing.

8 — Kill Switches

AP10 closes three kill switches (KS-2c, KS-15/D.2, KS-16) and engages three new ones (KS-D.1, KS-D.3, KS-D.4). The closures are major: the dimensional count of physical space, the uniqueness of the axiom-to-direction assignment, and the completeness of the four-axiom system. The live kill switches target the load-bearing structural identifications the argument relies on.

Here is how each closure was achieved, and here is where the live tests sit.

KS-2c — N = 3 spatial dimensions

Claim. The number of spatial dimensions is N = 3, derived from the axioms of the record algebra rather than imposed as an empirical fact.

Test. Demonstrate that N differs from 3 (the manifold has 2, 4, or some other number of spatial dimensions) and is not derivable from the four independent axioms. Empirical handles: high-energy physics signatures sensitive to extra dimensions (Kaluza-Klein modes, gravitational deviations from the inverse-square law at short scales), cosmological observations sensitive to dimensional count. Structural handles: an argument that the four axioms produce a different dimension count, or that a fifth independent axiom can be added.

Status. CLOSED — contingent on KS-D.3. The argument in sections 2–3 derives N = 3 from the four independent axioms via the proven faithful embedding plus the dimension-minimal identification (each independent axiom contributes exactly one new direction). The dimension-minimal identification is the load-bearing additional claim tracked by KS-D.3 as a live structural test. If KS-D.3 fires at any axiom, KS-2c reopens; until KS-D.3 fires, KS-2c stands. Section 5 provides structural consistency from the residual face count (the six faces pair into the axis structure of 3+1 spacetime), but does not constitute an independent derivation of N = 3 spatial.

Recovery. If KS-D.3 fires at any axiom, the count N = 3 changes (an axiom expressing zero faces lowers it; an axiom expressing two faces raises it). The recovery path is to re-derive the dimensional count from the corrected face structure. The Lorentzian signature (section 4; fenced at KS-D.4) is preserved either way; only the number of (+) directions changes.

KS-15 / KS-D.2 — Axiom-to-dimension assignment unique

Claim. The assignment R → time (the temporal direction with signature −) and {C, S, B} → space (the three spatial directions with signature +) is the unique assignment consistent with the Lorentzian signature (section 4's structural argument; formal derivation fenced at KS-D.4).

Test. Demonstrate an alternative assignment (one of C, S, B serving as the temporal direction; or R serving as a spatial direction; or the assignment being underdetermined) that is consistent with the axioms and the Lorentzian signature.

Status. CLOSED — contingent on KS-D.4. R is the only axiom that introduces irreversibility — the monoid has no inverse. All other axioms are compatible with reversibility in their contributed direction (C is a symmetric speed limit; S is an involution σ with σ^2 = identity; B contributes a definite break direction without prohibiting spatial reversal). Only one axiom can give the (−) direction: R. Once R is assigned to time, the remaining three axioms are forced to give the three (+) directions. The assignment is unique.

Recovery. N/A — closed, contingent on KS-D.4. The closure depends on R being the only irreversible axiom. If a structural argument were exhibited that one of {C, S, B} also implies irreversibility, the uniqueness would fail; in that case the recovery is to re-derive the irreversibility structure of the axioms and re-examine which axiom carries the (−) signature.

KS-16 — Sufficiency and exhaustiveness of {S, B, R, C}

Claim. The four canonical axioms {S, B, R, C} are sufficient to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1). The structural feature that might be expected as a fifth dimension is in fact the 1:1 itself — the pre-state, the probability space, the Hilbert space — which is prior to the manifold rather than a fifth direction on it. The exhaustiveness claim (no fifth independent axiom is possible) is held as a positive enumeration of operations on the 1:1: the four axioms address the four cardinal structural questions (distinguish via S, break via B, record via R, bound via C); no fifth cardinal structural question remains.

Test. For *sufficiency*: demonstrate a structural feature of the record algebra not derivable from {S, B, R, C}. For *exhaustiveness*: exhibit a fifth independent cardinal structural operation on the 1:1 that is not derivable from {S, B, R, C} and not subsumed under one of the four cardinal questions. Alternative: exhibit an observable feature of physical reality that requires a fifth manifold direction (a fifth spatial dimension empirically observed) or that the Hilbert space identification fails.

Status. CLOSED — contingent on KS-P.1. The sufficiency closure rests on the completeness result stated at AP20 section 4.1 (proof sketch carried there); its full formalisation is the obligation KS-P.1 names — if that obligation fails, KS-16 reopens in the sufficiency sense. The exhaustiveness sense rests on the positive enumeration argument in section 7.4: the four axioms address the four cardinal structural questions, and the 1:1 itself is the substrate, not an operation. The argument is structural rather than a closed theorem; if a fifth cardinal structural question is identified, KS-16 reopens in the exhaustiveness sense.

Recovery. If sufficiency is challenged, the recovery path is to identify which structural feature is missing from {S, B, R, C} and either derive it or add a fifth axiom; the completeness result at AP20 section 4.1 would fall (KS-P.1 fires) and the architecture would need to be re-derived from the expanded axiom set. If exhaustiveness is challenged (a fifth cardinal question is identified), the architecture would need to add a fifth axiom addressing that question; the dimensional count N = 3 would change accordingly via KS-D.3 (one axiom, one face); the Hilbert space identification of the 1:1 would need to be re-examined to see whether the new operation acts on the 1:1 as substrate or alongside it.

KS-D.1 — Six-face count and axis-structure interpretation

Claim. The multidimensional residual of the break has exactly six faces (AP06 section 10.5), and the six pair into the axis structure of 3+1 spacetime: two purely spatial axes (G–c, α/β–α_em) and one spatial-temporal connection axis (m_e–t).

Test. Demonstrate that the residual has a different number of faces (four, eight, ten); that the six identified faces do not pair into three conjugate pairs; or that the pairs do not correspond to the expected axis structure of 3+1 spacetime (e.g., the pairs would force a different N, or the temporal face t would have to be paired with another temporal face that does not exist in AP06's inventory). Specific structural handles: re-examine AP06's identification of {G, c, α/β, α_{em}, m_e, t} as the complete inventory; check whether any face has been missed, double-counted, wrongly paired, or wrongly interpreted as spatial-temporal vs purely spatial.

Status. LIVE — STRUCTURAL. The argument depends on AP06's residual analysis being exhaustive and correctly paired *and* on the pairs being correctly interpreted in the 3+1 axis structure. KS-D.1 is the structural test on all three dependencies.

Recovery. If KS-D.1 fires at the count or pairing level, the section 5 consistency check fails: a different face count or different pairing would not match the 3+1 axis structure, and the residual-side consistency would be lost. If KS-D.1 fires at the interpretation level (the pairs do not correspond to the expected axis structure), section 5 would still need to be reframed. In either case, the primary argument (sections 2–3) survives, as it is independent of the residual structure. Targets: section 5's consistency check; AP06 section 10.5's residual analysis.

KS-D.3 — One axiom, one face (dimension-minimal embedding)

Claim. Each of the four independent axioms expresses exactly one face of the embedded manifold — not zero, not two. The identity rests on the proven faithful embedding (AP20) plus the dimension-minimal principle: the embedding is minimal in dimension, so each independent algebraic axiom contributes exactly one new direction to the carrier manifold rather than zero (acting only as a constraint within existing dimensions) or two (contributing more than one independent direction).

Test. Demonstrate that an axiom expresses zero faces (acts only as a constraint within existing dimensions rather than expressing a new direction) or two faces (contributes more than one independent direction to the manifold). Or: demonstrate that the dimension-minimal principle fails — that the embedding is not dimension-minimal and the manifold can have more dimensions than the count of independent axioms would suggest. Specific structural handles: re-examine the section 2 argument for each axiom (R, C, S, B) and check whether the face-identification is precise. The two most exposed identifications are S → spatial face (a $\mathbb{Z}_2$ involution must give a continuous spatial dimension rather than a discrete internal index) and B → spatial face (the locus of ε must give a new spatial direction rather than a constraint within R+C+S). Both rest on the dimension-minimal principle applied at the specific axiom; both are flagged explicitly in section 2.

Status. LIVE — STRUCTURAL. The argument depends on the one-to-one correspondence between algebraic axioms and geometric faces, which in turn rests on the faithful embedding (proven, AP20) plus the dimension-minimal principle (the additional structural claim). KS-D.3 is the structural test on the central identity.

Recovery. If KS-D.3 fires, the count N = 3 is wrong. An axiom contributing zero faces would lower the count (the failed-axiom face count is zero, so the manifold has fewer dimensions than the count of axioms suggests); an axiom contributing two faces would raise the count. The recovery path is to re-derive the dimensional count from the corrected face structure. The Lorentzian signature (section 4; fenced at KS-D.4) is preserved either way; only the number of (+) directions changes. KS-2c is contingent on KS-D.3 remaining unfired and would reopen with the corrected count. Targets: section 2's axiom-to-face identification; the central identity *one axiom, one face* and the dimension-minimal principle on which it rests.

KS-D.4 — Signature formalisation

Claim. The embedded manifold carries a pseudo-Riemannian metric of Lorentzian signature (−, +, +, +). The structural argument is section 4's: exactly one axiom (R) is intrinsically irreversible — the monoid has no inverse — and irreversibility read as metric character is the (−) sign; the three reversible axioms carry (+). The formal metric-level derivation of this signature from the record algebra is the open obligation this switch names.

Test. Demonstrate that the embedded metric is not pseudo-Riemannian of signature (−, +, +, +) — for example, that the faithful embedding admits a Riemannian (all-+) metric consistent with the axioms; or that irreversibility of a generator does not force a (−) metric character; or that the formal derivation, once attempted, requires inputs beyond {S, B, R, C} and the proven EH.

Status. LIVE — STRUCTURAL. Section 4 carries the structural argument; the formal metric-level derivation is open. KS-15/D.2's closure (the uniqueness of the assignment) and the temporal-versus-spatial distinction of section 4 are contingent on this signature claim holding.

Recovery. If KS-D.4 fires at Lorentzian-ness itself, section 4's temporal-versus-spatial distinction fails: the architecture retreats to a four-dimensional manifold without metric character; the dimensional count (KS-2c) survives, and the assignment (KS-15/D.2) reopens. If it fires only at the formal level — the derivation requires an additional input — the input is declared and fenced, and section 4's structural argument stands as the conditional.

Why these six and not more

The six kill switches above cover the load-bearing claims of the paper. Each closure (or live test) addresses a structurally distinct concern.

KS-2c closes the central question of the paper — the dimensional count of physical space. KS-15/D.2 closes the uniqueness of the temporal-versus-spatial assignment. KS-16 closes the structural completeness of {S, B, R, C} by identifying the 1:1 as the Hilbert space rather than a missing fifth axiom. KS-D.1 tests the section 5 confirmation. KS-D.3 tests the central identity *one axiom, one face* that section 2 relies on. KS-D.4 tests the signature's formal derivation that section 4 relies on.

Two further structural claims that some readings would expect to carry their own formal kill switch are absorbed into the kill switches above. The faithful embedding (EH, AP20) is the foundation of section 3.2's argument that algebraic independence entails geometric face-independence; any failure of EH would falsify AP20, not AP10 specifically, and AP10's argument would then be untestable on the same manifold. The Lorentzian signature is carried in this paper: section 4's structural argument reads the (−) character from R's unique irreversibility, and the formal metric-level derivation is fenced at KS-D.4. EH is upstream of AP10 and tracked at its source paper; AP10 inherits that closure without re-engaging it.

The conjugate pair structure in section 5 (G–c, α/β–α_em, m_e–t) is held to formal kill-switch standard via KS-D.1 (the face count) without adding a separate kill switch for each pair: if the count and pairing both hold, the section 5 confirmation runs; if either fails, the confirmation fails. The two failures collapse into one test.

9 — What This Establishes and What Remains Open

9.1 — What this paper closes

Three kill switches, the central conditional in AP08's derivation of Einstein's field equations, and the completeness of the four-axiom system.

Specifically:

First. N = 3 spatial dimensions, derived from the four independent axioms of the record algebra via the proven faithful embedding plus the dimension-minimal principle (sections 2–3). The closure is contingent on KS-D.3 (one axiom, one face) remaining unfired; the residual-face structure of AP06 (section 5) is structurally consistent with this 3+1 derivation, providing a cross-check from a different feature of the architecture, but does not constitute an independent derivation of N = 3 spatial because Pair 3 includes the temporal face t.

Second. The axiom-to-direction assignment is unique. R is the only irreversible axiom; therefore R is the only axiom that can supply the (−) direction in the Lorentzian signature; therefore the assignment R → time, {C, S, B} → space is forced.

Third. The fifth structural degree of freedom is the 1:1 itself, not a fifth axiom. The 1:1 is what {S, B, R, C} act on; on the manifold, it appears as the Hilbert space — the probability dimension, the wave function's home. The four canonical axioms are sufficient to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1); the stronger exhaustiveness claim (no fifth axiom is *possible*) rests on the positive enumeration of operations on the 1:1 in section 7.4 — the four cardinal structural questions (distinguish, break, record, bound) are addressed by the four axioms, and the 1:1 itself is the substrate, not an operation.

Fourth. AP08's derivation of Einstein's field equations becomes unconditional. AP08 derives the equations via Lovelock's theorem conditional on N = 3; AP10 derives N = 3 from the axioms; the conditional is removed. Einstein's field equations with cosmological constant are an unconditional theorem of the record algebra.

Fifth. The 420 Code now derives, from the four axioms alone, Lorentzian spacetime, special relativity, N = 3 spatial dimensions, the Poisson equation, Einstein's field equations, the cosmological constant existence, and quantum mechanics. No empirical inputs other than m_e (from AP24) and the value of G (identified but not independently computed, per AP08 section 5). EH and QRA are proven (AP20). All results are unconditional.

9.2 — What this paper does not close

Two kill switches remain live, both structural rather than empirical.

KS-D.1 (Six-face count, LIVE — STRUCTURAL). The section 5 confirmation of N = 3 depends on AP06 section 10.5's identification of six residual faces. If the residual analysis is revised, the confirmation can be lost. The primary argument (sections 2–3) is independent of this and survives. Full statement at section 8.

KS-D.3 (One axiom, one face, LIVE — STRUCTURAL). The central identity *independent algebraic content = independent geometric face* rests on the proven faithful embedding (AP20). If any of the four axioms turns out to express zero or two geometric faces, the count N = 3 changes. Full statement at section 8.

Both live kill switches are structural rather than empirical because the question they ask is not whether the empirical world matches a prediction — N = 3 is already empirically confirmed — but whether the structural argument that derives N = 3 is correct in its identification of axioms with faces and faces with dimensions.

9.3 — Items held open without claiming debt status

One item is bracketed inside this paper.

The mapping from face to specific spatial coordinate. AP10 shows that there are three independent spatial faces (C, S, B), but the three spatial axes of physical space are interchangeable under spatial rotation (SO(3) symmetry). The argument does not specify which coordinate axis in physical space corresponds to which axiom's face. This is the expected outcome — spatial isotropy is a feature, not a bug — and held open without claiming a separate debt.

9.4 — Where the philosophical-register implementations live

AP10 is a structural derivation. Its philosophical-register companions — the chapters where the same structural facts are inhabited in the prose register of the Ø Models catalogue — are named here at the level of expected pairings; final placement is confirmed during the Ø Models compilation sweep alongside the Notebook II audit programme.

Dissolutions Ø (the chapter on dimensionality and the structure of space) carries the philosophical companion to sections 2–5: why three spatial dimensions, what the structure of space *is*, and how the question "why three?" dissolves once the four-axiom structure is in view.

Predictions Ø carries the empirical face of the closures: the consequences of $N = 3$ for Lovelock's theorem, Einstein's field equations, and the closure of the cosmological-constant derivation programme.

Resolutions Ø (the chapter on quantum mechanics and the Hilbert space) carries the philosophical companion to section 7: the 1:1 as the probability dimension, the canvas under the painting, the structural status of the pre-state.

The direction of acknowledgment is two-way. AP10 is the original derivation; the Ø Models chapters inhabit the result in the prose register. Final companion-pairing list is fixed at Ø Models compilation.

9.5 — Structural relationships to subsequent work

AP08 (The Identity) closes Notebook II. AP08's derivation of Einstein's field equations becomes unconditional with AP10 in place. The $N = 3$ conditional in AP08 section 9 is removed; Lovelock's theorem applies in 3+1 dimensions; $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ is the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives.

AP09 (The Break — QM) inherits the Hilbert space identification at section 7. AP09 derives quantum mechanics from the pre-state structure of the 1:1; AP10 identifies the 1:1 as the Hilbert space; AP09's pre-state formulation now reads against AP10's identification.

AP12 ($\hbar$ Identification) inherits the Hilbert space structure on which $\hbar$ operates. The Stone's-theorem-plus-Axiom-B forcing argument (AP12) acts on the probability dimension AP10 identifies.

AP19 (The Direction) is the source of the "three faces of one manifold" reading inherited by section 2.5 of this paper. The relationship is parallel rather than upstream/downstream: AP19 carries the structural reading; AP10 derives the dimensional count.

AP24 (The Residual) identifies m_e as the residual of the break, appearing in conjugate pair 3 of section 5. The residual identification is inherited from AP24; AP10 reads it as one of the six face-directions of three-dimensional space.

AP06 (The Leakage Constant) is the source of the multidimensional residual analysis used in section 5. The six-face count comes from AP06 section 10.5; AP10's reading of those six as three conjugate pairs is the structural content of section 5.

9.6 — Placement within The 420 Code

AP10 is the third chapter of Notebook II (Spacetime). The chapter sequence runs AP03 — AP05 — AP10 — AP08. AP03 derives the constants; AP05 walks the chain and reformulates the cosmological identity; AP10 pins the dimensional count; AP08 closes with the field equations.

AP10 sits at the structural centre of Notebook II for a specific reason: it is the paper that converts the manifold from "Lorentzian with undetermined N" (where AP01, AP20, AP03, AP05 left it) to "Lorentzian with N = 3 derived." Once AP10 has done its work, AP08's field-equation derivation has no remaining conditionals, and Notebook II reads as a closed structural chain from the four axioms to Einstein's equations.

Outside Notebook II, AP10 feeds the quantum mechanics programme via the Hilbert space identification (AP09, AP12) and the cosmological programme via the unconditional field equations (AP08 → cosmological constant → Predictions Ø). The fifth-degree-of-freedom identification in section 7 is one of the 420 Code's most structurally consequential moves: it explains both why no fifth spatial dimension is observed and why quantum mechanics has the Hilbert-space structure it has.

9.7 — How The 420 Code reads now

AP10 in place changes the reading of The 420 Code in four ways.

First. The dimensional count is no longer an empirical input. N = 3 is derived from the axioms in the same way the Lorentzian signature is derived from the axioms. The 420 Code does not import 3 from physics; it derives it.

Second. The Hilbert space has a structural origin. The probability dimension is the 1:1 itself — the pre-state, the substrate on which {S, B, R, C} operate. Quantum mechanics has the structure it has because the pre-state has the structure the 1:1 has. The wave function lives in the Hilbert space because the Hilbert space *is* the 1:1.

Third. Einstein's field equations are unconditional. With AP10's closure of the N = 3 conditional in AP08, Einstein's field equations with cosmological constant follow from the axioms alone via Lovelock's theorem in 3+1 dimensions. The cosmological constant is no longer a free parameter or empirical input at the structural level; its existence is a theorem.

Fourth. The sufficiency of {S, B, R, C} is established (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1). The four canonical axioms generate the record algebra; no fifth axiom is needed. The stronger exhaustiveness claim — no fifth axiom is *possible* — rests on section 7.4's positive enumeration: the four axioms address the four cardinal structural questions (distinguish, break, record, bound); no fifth cardinal question remains; the 1:1 itself is not an operation but the substrate operations act on. The architecture is closed at the level of the axioms in the sufficiency sense and structurally argued for closure at the level of exhaustiveness.

These four structural facts are what AP10 contributes to the reading of The 420 Code. They are inherited by every subsequent chapter that touches dimensionality, Hilbert space structure, field equations, or the completeness of the axiom system.

9.8 — The closing

You have walked through this paper. You have seen the four axioms named, the four faces identified, the independence of the faces derived from the independence of the axioms, the temporal versus spatial character assigned, the six faces of the residual counted as three conjugate pairs, the consequences for Einstein's field equations and quantum mechanics named, and the fifth degree of freedom located not as a missing axiom but as the 1:1 itself — the canvas under the painting, the probability space, the Hilbert space.

The universe has three spatial dimensions.

Not because that is what we measure.

Because the 1:1 has four operations performed on it, one of which is irreversible (giving time) and three of which are reversible (giving space).

There is no fifth spatial dimension because there is no fifth operation.

There is no fifth operation because the 1:1 is already fully broken by {S, B, R, C}.

And the 1:1 itself — the thing being broken — is the Hilbert space, the probability dimension, the wave function's home.

It is the fifth degree of freedom.

It was always there.

It was never missing.

You are looking at the complete structure.

Claim Summary

Section 0 (Dependency and Scope). Contract with the reader. Dependencies (AP20 for the canonical axioms — completeness/minimality obligations fenced at KS-P.1/KS-P.2 — and for the EH and QRA proofs; the Lorentzian signature carried at section 4, fenced at KS-D.4; AP06 section 10.5 for the multidimensional residual; AP08 section 9 for Lovelock; AP19 sections 2–3 for the three-faces reading), axiom mapping (R ↔ time; C, S, B ↔ three spatial faces; 1:1 ↔ the Hilbert space), epistemic status per section, kill switch summary, structural debts (none registry-level; KS-D.1 and KS-D.3 as live structural tests), items held open, structural relationships.

Section 1 (The Starting Point). FOUNDATIONAL. The four canonical axioms {S, B, R, C} stated. EH and QRA named as proven (AP20). Lorentzian manifold (M, g) with signature (−, +, …, +) and symmetry group SO(1, N) — the signature carried by section 4’s structural argument, formal derivation fenced at KS-D.4. N undetermined; this paper determines it.

Section 2 (Four Axioms, Four Faces). STRUCTURAL ARGUMENT. Each axiom expresses one independent face of the manifold. R → time (the irreversible face, signature −). C → propagation (the speed-limit face, signature +). S → exchange (the sector-crossing face, signature +). B → break (the locus-of-ε face, signature +). The four faces co-arise; the manifold is one structure with four faces, not four structures assembled together.

Section 3 (Independence). DERIVATION. The independence of the four axioms (the removal arguments at AP20 section 4.2, fenced at KS-P.2) combined with the proven faithful embedding (AP20) and the dimension-minimal principle yields the independence of the four geometric faces. Four independent faces yield four dimensions. Signature (−, +, +, +), dimension 3+1. **N = 3.** Two questions answered: not more than 3, because no fifth independent axiom is *needed* (the completeness result at AP20 section 4.1, fenced at KS-P.1 — sufficiency); not fewer than 3, because all four axioms are independent and each contributes one face under the dimension-minimal principle (live structural test, KS-D.3).

Section 4 (Character of Each Direction). DERIVATION. The (−) direction comes from the only irreversible axiom: R. The three (+) directions come from the three reversible axioms: C, S, B. The signature (−, +, +, +) is the axiom structure (R, C, S, B) read as metric character. Uniquely determined.

Section 5 (The Six Faces). STRUCTURAL CONSISTENCY CHECK. AP06 section 10.5 identifies six residual faces: G, c, α/β, α_em, m_e, t. These pair into the axis structure of 3+1 spacetime: two purely spatial axes (G–c, α/β–α_em) and one spatial-temporal connection axis (m_e–t). The pairing is structurally consistent with the 3+1 count derived in sections 2–3, providing a cross-check from a different feature of the architecture (the residual rather than the axiom count). Not an independent derivation of N = 3 spatial, because Pair 3 explicitly includes the temporal face t.

Section 6 (Consequences). SYNTHESIS + CLOSURE. KS-2c closed (contingent on KS-D.3). Lovelock's theorem (AP08 section 9) becomes unconditional. The chain from axioms to Einstein's field equations is closed: axioms → Lorentzian manifold → N = 3 → Poisson on M → Lovelock 1971 in general N, with 1972 uniqueness in 3+1 specifically → $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}$. No empirical inputs other than m_e and the value of G.

Section 7 (The Fifth Degree of Freedom). STRUCTURAL ARGUMENT + CLOSURE. The 1:1 itself is the fifth structural degree of freedom, not a fifth axiom. On the manifold it appears as the Hilbert space — the probability dimension, pre-spatial and pre-temporal. The four axioms {S, B, R, C} suffice to generate the record algebra (stated at AP20 section 4.1, fenced at KS-P.1; KS-16 closed contingent on KS-P.1). The stronger exhaustiveness claim rests on the positive enumeration of operations on the 1:1: the four axioms address the four cardinal structural questions (distinguish, break, record, bound); no fifth cardinal question remains.

Section 8 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-2c (N = 3, CLOSED — contingent on KS-D.3). KS-15/D.2 (axiom-to-dimension assignment unique, CLOSED — contingent on KS-D.4). KS-16 (sufficiency of {S, B, R, C}, CLOSED — contingent on KS-P.1; exhaustiveness held as positive enumeration). KS-D.1 (six-face count and axis-structure interpretation, LIVE — STRUCTURAL). KS-D.3 (one axiom, one face / dimension-minimal embedding, LIVE — STRUCTURAL).

Section 9 (What This Establishes and What Remains Open). SYNTHESIS. Five structural closures named (N = 3; assignment unique; fifth DOF as Hilbert space; Lovelock unconditional; the 420 Code chain to Einstein's equations complete). Two live kill switches as structural tests. One item held open (rotational-symmetry interchangeability of the three spatial axes). Placement within The 420 Code: third chapter of Notebook II (Spacetime). Closing on the complete structure.

Conditionality Footer

Dependencies

AP20 (the canonical axioms: formal statement section 3.3; completeness and minimality stated at section 4, fenced at KS-P.1/KS-P.2) — the floor of section 1 and the basis of section 3.

The Lorentzian signature (section 4's structural argument; formal derivation fenced at KS-D.4) — the signature with N undetermined that this paper carries and pins.

AP20 (EH and QRA proven) — EH invoked at section 3.2 to convert algebraic independence into geometric face-independence.

AP06 section 10.5 (multidimensional residual) — the six-face structure read at section 5.

AP08 section 9 (Lovelock's theorem, formerly conditional on N = 3) — AP10 removes the conditional.

AP19 sections 2–3 (three faces of one manifold) — the co-arising reading inherited at section 2.5.

Dependents

AP08 (The Identity) — Einstein's field equations become unconditional once AP10 removes the N = 3 conditional.

AP09 (The Break — QM) — inherits the Hilbert space identification at section 7 as the pre-state from which quantum mechanics operates.

AP12 ($\hbar$ Identification) — $\hbar$ operates on the Hilbert space identified by AP10.

AP24 (The Residual) — m_e identification reads against the residual structure used in section 5.

Predictions Ø — cosmological-constant existence and the unconditional Einstein's field equations feed downstream cosmological predictions.

Dissolutions Ø — the philosophical-register chapter on dimensionality inherits the “why three” closure.

Resolutions Ø — the philosophical-register chapter on quantum mechanics inherits the Hilbert-space-as-1:1 identification.

Kill switches closed by this paper

KS-2c. N = 3 spatial dimensions, derived in sections 2–3 via the proven faithful embedding plus the dimension-minimal principle. CLOSED — contingent on KS-D.3 (one axiom, one face) remaining unfired. Section 5 provides structural consistency from the residual face count, not an independent derivation.

KS-15/KS-D.2. Axiom-to-dimension assignment unique (R → time, {C, S, B} → space). Closed in section 4 — contingent on KS-D.4 (signature formalisation).

KS-16. Sufficiency of {S, B, R, C}; fifth degree of freedom identified as the Hilbert space. Closed in section 7 — sufficiency contingent on KS-P.1 (the completeness result stated at AP20 section 4.1). Exhaustiveness held as positive enumeration in section 7.4 — the four cardinal structural questions (distinguish, break, record, bound) are addressed by the four axioms.

Kill switches that remain live

KS-D.1 (LIVE — STRUCTURAL). Six-face count. See section 8 for the full register.

KS-D.3 (LIVE — STRUCTURAL). One axiom, one face. See section 8 for the full register.

KS-D.4 (LIVE — STRUCTURAL). Signature formalisation. See section 8 for the full register.

Structural debts owed

No registry-level structural debts are owed by AP10. The three live kill switches (KS-D.1, KS-D.3, and KS-D.4) are structural tests of load-bearing identifications, not unresolved derivations. The dependencies in section 0.2 are either proven (EH and QRA, AP20), stated and fenced (the axiom-set obligations, KS-P.1/KS-

P.2), or named as the structural input on which the argument rests (AP06's six-face count).

Items held open without claiming debt status

The mapping from face to specific spatial coordinate. The three spatial axes are interchangeable under spatial rotation (SO(3) symmetry of three-space); AP10 fixes the count and signature without fixing a coordinate-level identification. This is the expected outcome — spatial isotropy is a feature of three-space, not a gap in the derivation. Held open at section 0.7 and section 9.3.

References

Lovelock's theorem and the dimensional-count literature. References are sparse by design: the structural content of AP10 comes from the four axioms and their independence arguments (AP20 section 4.2, fenced at KS-P.2), not from the literature framework. The literature provides the formal vocabulary in which the structural facts are most naturally stated.

Lovelock, D. (1971). *The Einstein tensor and its generalizations.* Journal of Mathematical Physics 12, 498–501. The theorem invoked at section 6.2: in four spacetime dimensions (3+1), the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$. With N = 3 derived in this paper, Lovelock's theorem applies unconditionally to the manifold of the record algebra.

Lovelock, D. (1972). *The four-dimensionality of space and the Einstein tensor.* Journal of Mathematical Physics 13, 874–876. The dimensional uniqueness result that makes Lovelock's theorem particularly powerful in 3+1 dimensions: the divergence-free symmetric rank-2 tensor is unique in N = 3 spatial dimensions in a way that is not preserved in higher-dimensional generalisations.

Kantowski, R., and Sachs, R. K. (1966). *Some spatially homogeneous anisotropic relativistic cosmological models.* Journal of Mathematical Physics 7, 443. The anisotropic cosmological geometries on which the dimensional structure derived here is realised; referenced for completeness with AP05's junction formulation.

Internal references to The 420 Code (AP01, AP03, AP05, AP06, AP08, AP09, AP12, AP19, AP20, AP24, Dissolutions Ø, Predictions Ø, Resolutions Ø) are catalogued in the Conditionality Footer above. The 420 Code is published copyleft and free at the420code.org; the body of work links to all referenced internal papers.

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